

7 (a) force due to E -field is equal and opposite to force due to B -field
 $Eq = Bqv$
 $v = E/B$ B1
 B1
 B1 [3]

(b) *either* charge and mass are not involved in the equation in (a)
or F_E and F_B are both doubled
or E , B and v do not change M1
 so no deviation A1 [2]

8 (a) minimum frequency for electron to be emitted (from surface) M1